More Induction

WTP $\forall n \in \mathbb{Z}_{>0}, P(n)$

BC: Prove $P(1)$.

IS: Prove "$P(a) \Rightarrow P(a+1)$"

Prove: $\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Proof: Proceed by induction.

Base case: $P(1)$ is "$1^2 = 1(2)(3)/6$"

FE $1 = 1$.

Inductive step: Assume $P(a)$, i.e.,

$$1^2+2^2+\cdots+a^2 = a(a+1)(2a+1)/6.$$ (WTP $P(a) \Rightarrow P(a+1)$)

Adding $(a+1)^2$ gives

$$1^2+2^2+\cdots+a^2+(a+1)^2 = \frac{a(a+1)(2a+1)}{6}+(a+1)^2.$$

The LHS is the LHS of $P(a+1)$.

The RHS is $\frac{a(a+1)(2a+1)}{6} + (a+1)^2 =$

$$(a+1)\left(\frac{a(2a+1)}{6} + \frac{6(a+1)}{6}\right) =$$

$$(a+1)\left(\frac{2a^2+a+6a+6}{6}\right) = (a+1)\left(\frac{2a^2+7a+6}{6}\right)$$

$$= \frac{(a+1)(a+2)(2a+3)}{6} = \frac{(a+1)((a+1)+1)(2(a+1)+1)}{6}.$$

This is the RMS of $P(a+1)$. ☒

Claim: $\forall a \in \mathbb{Z}_{\geq 0}$, 3^a is odd.

Proof: BC: $P(0)$ is "3^0 is odd", i.e., "1 is odd". This is true.

IS: Assume $P(a)$, i.e., 3^a is odd.

(WTP: $P(a+1)$, i.e. 3^{a+1} is odd)

$$\overset{\shortparallel}{\boxed{3^a}} \cdot 3$$

Then $3^{a+1} = 3^a \cdot 3$. Since 3^a is odd & 3 is odd, and since the product of odd integers is odd, 3^{a+1} is odd. $\blacksquare$

Slogan: "Any time something "works" for 2 things, it works for many things".

Example: we know that "odd · odd = odd"

Pf: $\left((2a+1)\cdot(2b+1) = 4ab + 2(a+b) + 1 \right)$

rem-of 1.

Claim: If $a_1, \ldots, a_n$ are odd integers then $a_1 \cdot a_2 \cdot \cdots \cdot a_n$ is odd. $P(n)$

Pf: Proceed by induction.

BC is $P(1)$ i.e., "if a_1 is odd, then a_1 is odd". This is true.

$P(2)$ is "if a_1, a_2 are odd, then $a_1 a_2$ is odd." This is true and we previously proved it.

IS: Prove that for $n \geq 2$, $P(n) \Rightarrow P(n+1)$

Assume $P(n)$, i.e., "if $a_1, \ldots, a_n$ are odd then $a_1 \cdots a_n$ is odd".

(WTP: $P(n+1)$ i.e. ["if $a_1, \ldots, a_{n+1}$ are odd" then $a_1 \cdots a_{n+1}$ is odd],

Assume $a_1, \ldots, a_{n+1}$ are odd.

Then $a_1 \cdots a_n$ is odd because $P(n)$ is true. Then $(a_1 \cdots a_{n+1}) = (a_1 \cdots a_n) \cdot a_{n+1}$.

Since $(a_1 \cdots a_n)$ is odd and a_{n+1} is odd, since we know $P(2)$, their product is odd. Thus $a_1 \cdots a_{n+1}$ is odd $\blacksquare$

$$a+b = b+a$$

$$a+b+c = a+(b+c) = a+(c+b)$$
$$= (c+b)+a$$

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2 base cases

$$P(1) \wedge P(2) \wedge \big(P(n) \Rightarrow P(n+1) \big)$$

The proof that $P(n) \Rightarrow P(n+1)$

doesn't work for $n=1$.

Prove: $\boxed{n! > 2^n}$ for $n \geq 4.$

$P(n)$

$$n! = n \cdot (n-1)(n-2) \cdots 1$$

$n =$	$P(n) =$
1	$1! = 1 > 2$
2	$2! = 2 > 4$
3	$3! = 6 > 8$
4	$4! = 24 > 16$

Proof: Proceed by induction.

BC: $P(4)$ is "$4! > 2^4$", i.e., "$24 > 16$".
This is true.

Assume $n \geq 4$ and assume $P(n)$, i.e., $n! > 2^n$.

(WTP $P(n+1)$, i.e., $(n+1)! > 2^{n+1}$)

Multiplying $P(n)$ by $n+1$ gives

$$(n+1)\,n! > (n+1)\,2^n.$$

The RHS of this is $(n+1)!$.

Since $n \geq 4$, $n+1 \geq 5 \geq 2$.

Thus $$(n+1)\,2^n \geq 2 \cdot 2^n = 2^{n+1}.$$

We conclude that $(n+1)! > 2^{n+1}$ ∎

Fibonacci #'s

$F_1 = 1$, $F_2 = 1$, $F_3 = 1+1$

$F_4 = 3$, ...

Last perfect power

$1, 1, 2, 3, 5, \boxed{8}, 13, 21, 34, 55, 89, \boxed{144}, \ldots$

Defn: $\boxed{\begin{cases} F_1 = 1, \; F_2 = 1 \\ F_n = F_{n-1} + F_{n-2} \end{cases}}$

Same as

$$F_{n+1} = F_n + F_{n-1}$$

$$F_a = F_{a-1} + F_{a-2}$$

$$F_{n+2} = F_{n+1} + F_n$$

$$F_{2n+2} = F_{2n+1} + F_{2n}$$

Claim: "$F_1 + F_3 + F_5 + \cdots + F_{2n-1} = F_{2n}$"

"$P(n)$"

Ex's 1, 1, 2, 3, 5, 8, 13, 21

Proof: (BC) $P(1)$ is "$F_1 = F_2$", i.e.,

"$1 = 1$". This is true.

IS: Assume $P(n)$, $n \geq 1$

"$F_1 + F_3 + \dots + F_{2n-1} = F_{2n}$".

$2(n+1)-1 =$
$2n+2-1$
$2n+1$

Adding F_{2n+1} to both sides gives

$$F_1 + F_3 + \dots + F_{2n-1} + F_{2n+1} = F_{2n} + F_{2n+1}.$$

By the defn, $F_{2n} + F_{2n+1} = F_{2n+2}$.

Thus $F_1 + F_3 + \cdots + F_{2n+1} = F_{2n+2}$, i.e.,

$P(n+1)$ is true. ∎

"Lucas"

$2, 1, 3, 4, 7, 11, 18, 29, \ldots$

$L_1 = 2 \qquad L_2 = 1$

$L_n = L_{n-1} + L_{n-2}$

<u>Claim:</u> $\forall n \in \mathbb{Z}_{>0}$, "$F_n < 2^n$".

<u>Proof:</u> (BC) $P(1)$ is "$F_1 < 2^1$", i.e., $1 < 2$. $P(2)$ is "$F_2 < 2^2$", i.e., $1 < 4$. These are true.

IS: Assume $P(n)$ and $P(n+1)$, I.E.,

$F_n < 2^n$ and $F_{n+1} < 2^{n+1}$.

(WTP $P(n+2)$, i.e., $F_{n+2} < 2^{n+2}$)

By defn $F_{n+2} = F_{n+1} + F_n$. Since $P(n)$ and

$P(n+1)$ are true, $F_{n+1} + F_n < 2^{n+1} + 2^n$, IH, $F_{n+2} < 2^{n+1} + 2^n$. (want 2^{n+2})

Since $2^n < 2^{n+1}$, $2^{n+1} + 2^n < 2^{n+1} + 2^{n+1}$

$= 2 \cdot 2^{n+1} = 2^{n+2}$. Thus

$F_{n+2} < 2^{n+2}$. $\blacksquare$

$$P(1) \wedge P(2) \wedge \big(P(n) \wedge P(n+1) \Rightarrow P(n+2) \big)$$

$$P(1),\ P(2),\ P(3),\ P(4),\ P(5),\ \ldots$$

Claim: "$F_{n-1} \cdot F_{n+1} = F_n^2 + (-1)^n$" $= P(n)$

1, 1, 2, 3, 5, 8

Proof: "$P(2)$" is the statement

"$F_1 \cdot F_3 = F_2^2 + (-1)^2$", i.e.,

$$1 \cdot 2 = 1^2 + (-1)^2$$

$2 = 2$. This is true.

IS: Assume P(a), i.e., $F_{a-1} \cdot F_{a+1} = F_a^2 + (-1)^a$.

(WTP P(a+1), i.e., $F_a \cdot F_{a+2} = F_{a+1}^2 + (-1)^{a+1}$)

By defn, $F_{a-1} + F_a = F_{a+1}$, i.e., $F_{a-1} = F_{a+1} - F_a$.

Subing gives $(F_{a+1} - F_a) \cdot F_{a+1} = F_a^2 + (-1)^a$.

Then $F_{a+1}^2 - F_a \cdot F_{a+1} = F_a^2 + (-1)^a$.

then $F_{a+1}^2 - (-1)^a = F_a^2 + F_a \cdot F_{a+1}$.

Then $F_{a+1}^2 + (-1)(-1)^a = F_{a+1}^2 + (-1)^{a+1}$

$= F_a(\underline{F_a + F_{a+1}}) = F_a \underline{F_{a+2}}$ by the def. $\blacksquare$

def of fib.